

Opinion

How useful are thermal vulnerability indices?

Susana Clusella-Trullas,^{1,*} Raquel A. Garcia,¹ John S. Terblanche,² and Ary A. Hoffmann³

To forecast climate change impacts across habitats or taxa, thermal vulnerability indices (e.g., safety margins and warming tolerances) are growing in popularity. Here, we present their history, context, formulation, and current applications. We highlight discrepancies in terminology and usage, and we draw attention to key assumptions underpinning the main indices and to their ecological and evolutionary relevance. In the process, we flag biases influencing these indices that are not always evaluated. These biases affect both components of index formulations, namely: (i) the characterisation of the thermal environment; and (ii) an organism's physiological and behavioural responses to more frequent and severe warming. Presently, many outstanding questions weaken a thermal vulnerability index approach. We describe ways to validate vulnerability index applications and outline issues to be considered in further developing these indices.

Vulnerability indices to climate change

As climate continues to change at an unprecedented rate, humans are witnessing impacts on multiple dimensions of biodiversity [2,3]. Climate change can directly and indirectly impact natural and managed ecosystems via alterations of population phenology, rates of extinction and proliferation, affecting human and animal well-being, with ectotherms expected to be most affected [3]. Consequently, there is a pressing need to forecast the potential impacts of climate change, by knowing when and where to focus management efforts [3,4].

Methods used to assess species' vulnerability to climate change include correlative, mechanistic and trait-based approaches or a combination of these [4]. Recently, simple vulnerability indices based on species' thermal performance, referred to as **thermal safety margin (TSM)** and **warming tolerance (WT)** [1] (see Glossary), have increasingly been used for ectotherms (111 articles between 2008 and 2021; see also supplemental material online). These indices are based on the direct influence of climate variables, mostly temperature, on the performance of organisms and they measure the extent to which performance optima or temperature thresholds such as critical limits are likely to be approached or exceeded with climate warming. The premise for such an approach is that the temperature sensitivity of performance (or physiology and fitness) mediates species distributions and persistence [5,6] and that these indices provide an absolute measure of vulnerability for an organism (as a single estimate) or a relative measure across species or populations.

Statements about vulnerability and extinction probabilities continue to be based on these indices [2] despite some studies raising notes of caution. There is now more of a focus on how animals perceive and use their environments and how undersampling of microsites available in time and space increases uncertainty around these indices [7,8]. There is also recognition that behavioural buffering and adaptive potential alter relationships between environmental variables and species characteristics that affect vulnerability [9–11]. These characteristics can be evaluated through simple indices such as levels of genomic variation, heritability measures and **acclimation** capacity. However, it remains challenging to incorporate these multiple facets into meaningful

Highlights

Accurate forecasts of the impacts of climate change are essential for managing biodiversity and minimising further species losses. Indices reflecting forecasts are useful when they are simple enough to be implemented at large scales, but this needs to be balanced against their ability to predict actual impacts.

Thermal vulnerability indices such as safety margins and warming tolerance have increased in popularity for making both regional and global predictions of species vulnerability to climate change, but concerns have been raised about their forecasting potential.

By estimating the deviation between species' thermal sensitivity and thermal exposure, these indices often do not reflect variation relevant to the organism. They also fall short of considering many intrinsic and extrinsic factors that affect species vulnerability, weakening their usefulness.

Under specific circumstances, safety margins can however answer particular questions relating to species' relative sensitivity and exposure, and can be complemented by other approaches, all of which require validation.

¹Department of Botany & Zoology, Stellenbosch University, Stellenbosch, South Africa

²Department of Conservation Ecology & Entomology, Stellenbosch University, Stellenbosch, South Africa

³School of BioSciences, Bio21 Institute, University of Melbourne, Melbourne, Australia

*Correspondence: sct333@sun.ac.za (S. Clusella-Trullas).



indices when making broad scale comparisons. These acknowledged problems and their solutions mean that studies using thermal vulnerability indices can be limited in their explanatory power. This article aims to cover a brief history of the origin of vulnerability indices, describe their ecological meaning relative to the context and formulation, discuss their primary limitations and validity, and propose ways forward towards an integrated and multifaceted framework for assessing population and species vulnerability to climate change.

History of thermal vulnerability indices and inconsistencies

The terms safety margin and warming tolerance were applied by Deutsch and colleagues [1] and described as 'heuristic indicators of the impact of warming on the thermal performance of insects'. Warming tolerance was defined as 'the average amount of environmental warming an ectotherm can tolerate before performance drops to fatal levels', while the thermal safety margin was defined as 'the difference between an organism's thermal optimum and its current climate', reflecting how much buffering the animal has before temperatures are too high to maintain optimal performance. Since then, there has been confusion in the way these indices have been formulated and applied.

Out of 111 articles that implemented these indices since Deutsch *et al.* [1], 75% provided warming tolerance estimates, 7% presented safety margins, and 18% calculated both. Among those presenting warming tolerance, authors referred to this index as warming tolerance, safety margin, thermal buffer, or thermal allowance [9]. Coincidentally, these studies did not always use the formula of [1]. To complicate matters, in 1970, Heatwole had called physiological thermal safety margin the difference between the mean preferred body temperature (often used as a proxy for T_{opt}) and the CT_{max} [12]. He also defined the ecological or minimum thermal safety margin as the difference between the maximum field body temperature and CT_{max} . The latter provided an 'indication of minimum safety margin that buffers a species under a given set of conditions'. Therefore, warming tolerance indices that use maximum **operative temperature** (T_e) or body temperature (T_b) in current articles resemble Heatwole's minimum thermal safety margins.

In turn, the thermal safety margin defined as ($T_{opt} - T_{hab}$) resembles the thermal quality index used to calculate the effectiveness of thermoregulation in ectotherms [13]. This index is calculated as the average of absolute deviations between the operative temperature and the preferred body temperature. A large deviation reflects a habitat of low thermal quality because animals have to invest more time or energy to thermoregulate and attain preferred temperatures. By contrast, in the context of climate change, a large deviation is interpreted as providing more buffer from warming while small deviations will result in higher warming risk. The lack of consistent index terminology and formulations can create unreliable interpretations, make comparisons across studies difficult, and spread uncertainty among scientists, managers and policy makers. Given the mixed appellation of these indices and to avoid further confusion here, we adopt WT and TSM for warming tolerance and thermal safety margins, respectively [1]. We also place more emphasis on WT because the variation in critical temperature thresholds and its determinants are better known and studied in the literature than parameters related to optimal temperature.

Ecological significance of thermal vulnerability indices

Thermal vulnerability indices incorporate one aspect of a species' intrinsic sensitivity to temperature (e.g., CT_{max} or T_{opt}) and one aspect of its exposure to temperature in the environment (e.g. mean or maximum habitat or operative temperature). Given the potential variation in each side of this equation (Boxes 1 and 2), the interpretation of thermal vulnerability indices needs to be placed within the context of the question, the index formula, the spatial and temporal scale used, and the ecological system studied.

Glossary

Acclimation, acclimatisation: forms of phenotypic plasticity or genotype-by-environment interactions occurring at different timescales triggered by changes in environmental conditions (e.g., seasonal or extreme events) that alter thermal tolerance, performance, and behaviour of individuals. Rapid acclimation responses are also referred to as hardening.

Behavioural thermoregulation: behavioural responses (e.g., moving between microsites in the sun and shade) to achieve an optimal body temperature and avoid exposure to extreme conditions.

CT_{max} : a functional upper critical thermal limit, or maximal temperature before loss of activity. In many organisms, this may be assayed by scoring a behavioural or physiological stress trait (e.g., onset of panting, or loss of righting response) under controlled, warming conditions. Diverse traits and methodological approaches are often lumped under the same rubric of CT_{max} .

CT_{min} : lower critical thermal limit, or minimal temperature of voluntary activity. Markers of neuromuscular function are typical traits considered.

Operative temperature: an index that combines properties of the organism (such as size and shape) and the microenvironment (such as exposure to solar radiation and wind) to estimate the body temperature of an organism in equilibrium with its environment. It does not include metabolic heating or evaporative cooling. T_e is typically obtained with replicas of organisms with zero-heat capacity or through heat transfer equations.

Thermal safety margin: thermal vulnerability index defined as the difference between a particular trait's optimal or maximal performance temperature and environmental or operative temperature. Referred to as TSM following [1].

T_{opt} : optimal performance temperature, or the temperature at which performance or response rate is maximised. Traits such as sprint speed are typically measured at the scale of seconds to minutes, while fecundity and growth are measured over weeks or months. T_{opt} measures need to be connected to some measure of fitness.

Warming tolerance: thermal vulnerability index generally defined by

As an absolute measure for a species

By incorporating a single performance threshold, WT disregards other aspects contributing to species' sensitivity to temperature change (Figure 1). For example, additional physiological and ecological characteristics (behaviour, trophic specialisation, and biotic interactions), life history, and genetic diversity can contribute to adaptive responses and resilience of species to climate change [34]. Furthermore, TSM and WT provide an ecologically limited view of thermal exposure by incorporating arithmetic means or maximums of temperature as the sole determinants of climate change impacts (Box 1), but changes in thermal temporal and spatial variability, including seasonal variation and frequency of extreme events, are ubiquitous and expected to increase [35]. Importantly, statistical descriptors of past environments are used to predict vulnerability in the future, with typically poor acknowledgement of known uncertainty around changes in climate.

For species with small distribution ranges, poor dispersal ability, limited evolutionary and acclimation capacity, limited behavioural strategies and short life spans, or species with critical, sessile or heat-sensitive stages (e.g., egg stage), WT measured at appropriate life stages could be a crude, but perhaps acceptable, index of vulnerability (Table 1). However, sublethal effects may accumulate and elicit a greater impact eventually. Thermal vulnerability indices do not integrate different life stage vulnerabilities nor assess fitness consequences (Figure 1) or provide information on how the effects of warming accumulate despite compensatory processes to elicit negative population growth. It may be possible to measure sublethal effects in a standardised way as proposed for fertility [33] and thermal effects on growth [36]. However, in most cases, a challenge still resides on deciding the scale of T_e data to use as we do not know the capacity of species to adapt through evolutionary changes and phenotypic plasticity, or to track suitable T_e . Different T_e scales will be relevant to CT_{max} and T_{opt} and data such as body size, generation time, dispersal and thermoregulatory abilities, may guide the choice of scale (Table 1).

A relative measure within and between species

Typically, intraspecific studies compare relative vulnerabilities of populations among sites, along geographic or climate gradients, seasons or life stages (Table 1). Most studies use mean T_e or maximum

the difference between an animal's habitat or operative temperature and its upper thermal limit. The index varies markedly depending on the source and is here referred to WT as in [1].

Box 1. Variation in the thermal exposure side of the TSM and WT equations

The thermal exposure of an ectotherm is dictated by the operative temperatures available in its environment. Operative temperatures integrate properties such as body size, surface area, shape, and skin reflectance with pathways of energy from the environment (solar radiation, convection, and conduction) that are also influenced by abiotic and biotic factors (e.g., rock formations, soil properties, and crowding) [14,15]. The animal's behaviour (e.g., activity period, foraging mode, extent of behavioural thermoregulation, and habitat use) and dispersal abilities also shape the extent of exposure [15]. Therefore, the spatial and temporal sampling of operative temperatures needed to define thermal exposure depends on the organism and the study aims (Table 1). Many studies use environmental (air or water) temperature data at a coarse spatial scale, modelled from weather stations or remote sensing data. However, operative or body temperature measured in the field, more relevant to the study organism, are increasingly used (Figure S1 in the supplemental information online). Temporal extents and resolutions range from a single time point to multiple years and from hourly to monthly. High temporal resolution data at fine spatial scales are typically collected to the detriment of longer temporal extents, especially when using physical replicas of the organism to obtain T_e (Figure S1 in the supplemental information online). Generally, there is limited emphasis on choosing a sampling regime relevant to the organism (e.g., behaviour and body size) or to adaptive and evolutionary processes that shape its sensitivity parameters (T_{opt} or CT_{max} , Box 2).

Furthermore, the choice of statistic summarising the thermal environment (e.g., mean vs maximum) can drastically change the index value. The mean annual maximum temperature in the shrubland (2, orange T_e distribution) is below CT_{max} , whereas the daily maximum temperature (3) has already surpassed this physiological limit (Figure 1B). WT is higher in the shrub-land and desert than in the forest during mid-morning and mid-afternoon while it is largest for the tropical species at noon (Figure 1C). Therefore, vulnerability can be over- or underestimated depending on the sampling of T_e in the habitat both in space (B) and time (C) and the summary statistic used. Most studies use mean temperature for TSM and maximum values for WT, but there is variation affecting the accuracy and ecological meaning of indices, leading to disparate results and interpretations of vulnerability [7,16].

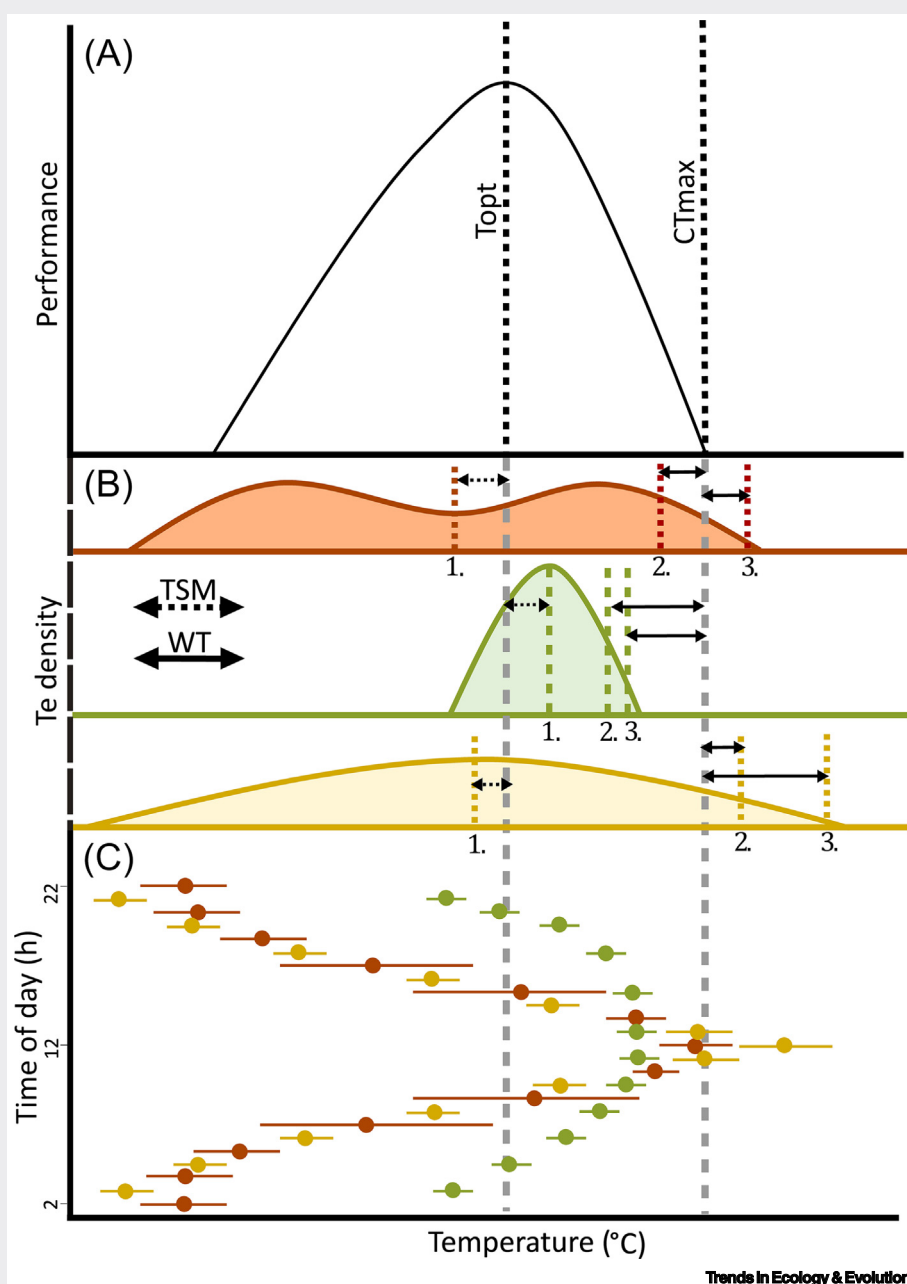
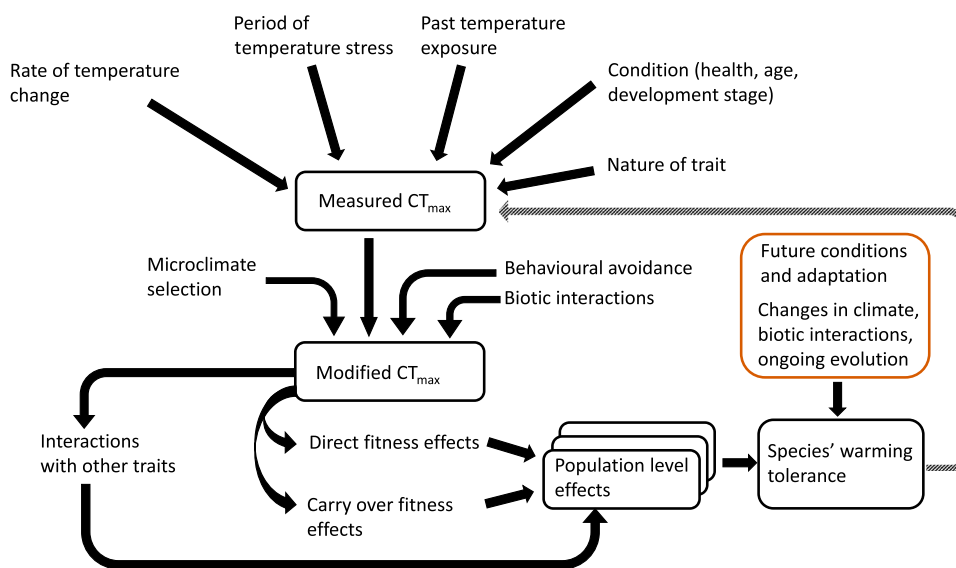


Figure 1. Potential effects of variation in operative temperature on thermal vulnerability indices. (A) A hypothetical thermal performance curve depicting the temperature at optimal performance (T_{opt}) and the critical thermal maximum (CT_{max}). (B) Examples of frequency distributions of operative temperatures (T_e) in three different habitat types (orange: temperate shrubland; green: tropical forest; and gold: desert). The vertical stippled lines numbered 1–3 represent summary statistics of environmental temperature: 1, mean T_e; 2, mean annual maximum of T_e; and 3, mean daily maximum of T_e. The stippled horizontal arrows are thermal safety margin (TSM) (T_{opt} minus mean T_e) while black horizontal arrows are warming tolerance (WT) (CT_{max} minus mean annual (2) or daily (3) maximum T_e). (C) The hypothetical density distributions in B are decomposed into bihourly T_e means (circles) and interquartile range (horizontal lines) across the day and for each habitat type. The vertical lines in A representing CT_{max}, T_{opt}, and CT_{min} are extended into lower panels to show how TSM and WT are influenced by habitat-specific T_e densities and summary statistics in B, and by the daily temperature time series in C.

Box 2. Variation in the species' sensitivity side of the WT and TSM equations

Upper thermal limits and optimal performance temperatures can vary substantially, both within and among ectotherms (e.g., for CT_{max} , $>25^{\circ}C$ variation globally across insects [17]; $>3^{\circ}C$ variation within lizard species [18,19]). This variation can be partitioned into signal, noise (e.g., random variability) and systematic methodological biases, or their combination. Within species, variation in upper thermal limits and optimal performance temperature associates with intrinsic (e.g., body size, age, and sex) and extrinsic (e.g., seasonality) factors that may promote phenotypic plasticity of performance parameters [20] (see Figure 1 in main text), although this will differ for CT_{max} and T_{opt} . Across species, key sources of variation in CT_{max} include rates of temperature change used during assays, adaptation to current climatic extremes and evolutionary constraints [17,21,22]. For T_{opt} , interspecific variation is typically less understood and is likely influenced by many factors, including the nature of the trait scored and how closely it translates to fitness [20]. Depending on the focus of a study, the signal might include plastic physiological or behavioural effects. In other cases, the noise might encompass plastic trait effects (e.g., seasonality and hardening).

Choice of experimental methodology can also mask (or reveal) variation in the signal of interest [23,24]. Long duration stress assays can produce lower estimates of CT_{max} compared to shorter duration stress assays, altering the shape of thermal performance curves, with implications for WTs [25]. Consequently, long duration tests of thermal sensitivity can be (falsely) associated with higher thermal sensitivity relative to short duration tests. To some extent, different metrics of heat stress resistance (e.g., CT_{max} and heat knockdown time, or dynamic versus static assays) can be interchangeable or mathematically related [26,27]. However, this is not always the case and CT_{max} assays mimicking diurnal heating rates can reveal genotypic (and phenotypic) capacity masked by other types of assays (e.g., acute stress) [28]. Less attention has been given to methodological artefacts in T_{opt} estimation, although this measure needs to be clearly linked to fitness measures and similar issues apply [29,30]. In addition, the shape and the descriptors of the thermal performance curve (CT_{max} and T_{opt}) can vary widely depending on the performance trait examined (e.g., running speed, rates of development, and reproduction) [29,31]. So far, performance traits used in WT estimations have primarily included locomotion and lethal thresholds, and a few have used intrinsic population growth rate, and cardiac or metabolic function. For locomotion, thresholds vary considerably within and between taxonomic groups, including onset of spasms, and loss of righting response, equilibrium, or muscle coordination. This variation can modify the ecological interpretation of WT but also increase or reduce a species' apparent vulnerability [32,33].



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Figure 1. Connections between CT_{max} measurements taken on individuals and WT of a species. Measured CT_{max} is influenced by a number of factors. Organisms will have their actual CT_{max} affected by the microclimate they select as well as behavioural and biotic factors. The impact of CT_{max} needs to be linked to fitness effects within and across generations, which will determine population level effects and the actual warming tolerance of populations and species under the complex effects of changing future conditions (orange box). The shaded arrow depicts how joined sensitivity and exposure effects can drive the evolution of thermal limits. Many of these factors will also be important for other measures such as T_{opt} although for T_{opt} the rate of temperature change may be less important. Abbreviations: CT_{max} upper critical thermal limit; WT, warming tolerance.

Table 1. Potential range of focal questions or research objectives that can be addressed using a thermal vulnerability index approach

Focal question/research objective	Suggested calculation or approach	Key considerations
A. How much buffering from warming does a species have on average? [63]	1. $\overline{CT_{max}} - \overline{T_e}$; 2. $\overline{T_{opt}} - \overline{T_e}$ Combining both and calculating the proportion of T_e or time with T_e above physiological thresholds can provide useful information. Index can include within-species variation, e.g., 3. $\frac{\sum_{i=1}^n (\overline{CT_{max,i}} - \overline{T_{e,i}})}{n}$ and s.d., s.e. or confidence limits. This formula can be applied to most equations in this table.	Only a single aspect of sensitivity (e.g., running speed performance, reproduction) is normally considered (but see question G) Causes of variation of CT_{max} (Figure 1) and T_{opt} should be considered to define appropriate T_e (e.g., max T_e may be more relevant for (acute stress) upper temperature thresholds while long-term mean T_e may be more relevant for T_{opt} and the critical limit for growth and reproduction).
B. How much buffering does a species have from extremes? [64]	4. $\overline{CT_{max}} - \overline{T_{e,max}}$ Parameters of $T_{e,max}$ include mean annual max, daily max, or can be tailored to encompass intensity and duration of extreme events [40,65].	The $T_{e,max}$ parameter should be calculated from T_e data with temporal and spatial scales relevant to the organism (life span, buffering capacity, body size, and dispersal).
C. How much buffering can a species reach given thermal refuges?	5. $\overline{CT_{max}} - \overline{T_{e,min}}$ The assumption is that the organism can use $T_{e,min}$ despite the need to forage, feed, and perform key activities. $T_{e,min}$ can be calculated from e.g. daily min T_e across available microsites.	As in B., life history, behaviour, and energetic requirements should be considered when determining the temporal and spatial scale used to obtain mean $T_{e,min}$.
D. How much buffering do species actually use? How much of this buffer is driven by T_e constraints?	6. $\overline{CT_{max}} - \overline{field\ T_b}$ This index can be combined with index 5 (above) to determine if the buffer is shaped by T_e constraints and/or organism's choice. Scatterplots of respective deviations can reveal equivalence when field T_b s match $T_{e,min}$.	The shape of T_b and T_e frequency distributions and the time series data should be examined.
E. How much does WT change when accounting for potential behavioural thermoregulation ? [9,64]	WTs using field T_b and mean $T_{e,max}$ and $T_{e,min}$ (formulas 4., 5., and 6. above) can be examined in combination.	T_e frequency distributions can be decomposed into relevant time series (Box 1) providing a better indication of T_e availability.
F. How does thermal vulnerability change with and without thermal plasticity? [10,39,66]	7. $\overline{CT_{max}} - \overline{T_{e,max}}$ versus $\overline{CT_{max,Acc}} - \overline{T_{e,max}}$ The left equation disregards acclimation while the right one includes a CT_{max} adjusted for plasticity. The adjustment can be determined empirically based on particular warming regime (e.g., acute versus chronic) or calculated based on the species' acclimation response ratio (change in thermal tolerance relative to acclimation temperature exposure) and the difference between experimental and $T_{e,max}$ (or temperature regime experienced prior to $T_{e,max}$ [66]).	Incorporate adequate environmental signals of acclimatisation and time scales of response of the organism [24,44].
G. How does thermal vulnerability vary depending on the trait measured (e.g., are WTs lower for fertility than for lethality)? [33,67]	7. $\overline{CT_{max,Trait1}} - \overline{T_{e,max}}$ versus $\overline{CT_{max,Trait2}} - \overline{T_{e,max}}$ Rather than a comparison, different trait WTs can be combined or integrated in multidimensional space (e.g., defining polygons or component of a multivariate analyses).	The nature of traits (e.g., lethal versus sublethal effects: CT_{max} vs locomotion, growth) and how these are determined should dictate the scale and period for T_e . When multiple traits are available, mechanistic (dynamic energy-budget) models could be implemented [68].
H. How much do thermal vulnerability estimates change depending on the nature and scale of environmental temperature (e.g. T_{air} , T_e , T_b ; fine, coarse gridded data)? [7,69]	Indices 1–4 are used to assess the effects of different types of temperature exposure parameter on index estimations. This approach can be used to assess biases originating from, for example, logistical constraints when measuring T_e . Sensitivity analyses can be included in these analyses.	Uncertainty around WT originating from T_e should be considered. The choice of scale and duration of T_e data should be shaped by factors relevant to the organism [e.g., response capacity of species (dispersal, thermoregulation, plasticity) and adaptive potential].
I. What is the thermal vulnerability of a species across time or space (e.g., across seasons, ontogenetic stages; latitudinal, aridity or urbanization gradients; distribution range)? [6,70,71]	Indices 1–4 can be used to compare vulnerability across time or space.	Appropriate scales of T_e need to be used depending on the study question and focal organism.

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Table 1. (continued)

Focal question/research objective	Suggested calculation or approach	Key considerations
J. What is the relative thermal vulnerability of populations, species or communities? [1,2,72,73]	Indices 1–4 can be used to compare populations or species. This approach can be used to compare indices in environments with different degrees of predation or competition, and predator–prey or invasive–native species counterparts. Community indices can include species richness or species relative abundance as weights to derive WT weighted means.	Appropriate scales of T_e need to be considered at each level of organization. Conditions and durations used to determine CT_{max} and T_{opt} can confound patterns across populations or species (Box 2).
K. How will the thermal vulnerability of species or communities change from current to future climate conditions? When in the future will current WTs be exceeded? [2,48,74,75]	$\frac{CT_{max_present} - \overline{T_{e_max_present}}}{CT_{max_future} - \overline{T_{e_max_future}}}$ versus $CT_{max_future} - \overline{T_{e_max_future}}$	Both the sensitivity and exposure sides of the WT equation should include variability in the present (i.e., plasticity and intraspecific variation for CT_{max} plus heterogeneity of T_e). In future scenarios, trait evolutionary potential and uncertainty around T_e should be considered.

We offer ways to improve index estimates to address specific study objectives and present limitations or aspects that need to be considered when making interpretations of thermal vulnerability indices. When available, we provide references that include a similar question and/or approach (note that most studies do not use T_e as recommended in the formulas).

T_e to calculate WT, with maximum values considered to be better indicators of heat stress or exposure and to contribute to the evolution of CT_{max} variation [37,38]. However, given intraspecific variation in the sensitivity and exposure determinants of the WT equation (Boxes 1 and 2), there will be biases in index estimates across populations. For example, plasticity of tolerance and availability of microsites can vary across seasons and populations, and this variation should be incorporated into index estimates (Table 1) [39]. Alternatively, when performance curves for multiple populations or time periods are available, integrating performance with temperatures in the field over a specific period can provide estimates of realized thermal stress (when performance <0) or of fitness decline across populations or through time (Table 1) [20,40,41].

For interspecific studies, most comparisons include taxa across large geographic scales. Most of the limitations highlighted for within-species comparisons are relevant here and mask responses across species. Moreover, while TSMs and WTs provide a simplistic but efficient way to compare vulnerability across species, they neglect to integrate cause and effect pathways that would allow scaling up from individuals to species and communities. An exception is heat waves causing local population extinctions, with a direct realised impact on species [42]. In these cases, adequate temporal and spatial sampling of temperature extremes relevant to species' life histories is required. Variation in the magnitude, timing, frequency, and duration of extreme events and the temperature experienced between them can also affect the degree of heat stress or injury and potential repair and recovery across species [30,43]. These balancing effects cannot be predicted when using summary statistics of T_e and require knowledge of long-term temperature series coupled with empirical data on species-specific responses (Table 1) [44]. In addition, the temperature range between CT_{max} and lethal thresholds varies greatly among taxonomic groups. For example, CT_{max} is typically fatal or near-fatal in most insects. This is not always the case for reptiles and amphibians unless temperature continues to increase or stressful conditions persist for long periods. Consequently, compilations of WT across species to infer fitness effects or climate vulnerability need to be mindful of this varying CT_{max} -performance-fitness relationship and sources of variation (Table 1) [25].

Assessing the validity of indices

There are several ways of validating vulnerability indices, linking data directly to changes in distributions, population dynamics and, ideally, likelihood of population extinction. These opportunities

for validation are similar to those involved in thermal limits [24]. One approach is to take situations where data are available for a group of species on WT, climate and distribution shifts, and examine links between them. WTs have so far not predicted range shifts for bumblebee species in Europe and North America [45], but thermal tolerances did predict the response of bees to urban heat islands [46] and of ant species to experimental warming at their southern range limit where TSMs were expected to be small [47]. There may be more success in aligning thermal vulnerability indices with range shifts in marine species, which experience lower temperature variability and have narrower tolerance limits than terrestrial species [6,48]. Extirpations linked to warming are more frequent in marine than terrestrial organisms, likely because terrestrial species have greater access to favourable microclimates, have wider WTs, greater ease in tracking suitable climate space and lower influence of temperature on range boundaries.

It may be possible to link population dynamics and thermal responses in other ways, such as through linking changes in population sizes of short-lived insects with environmental temperatures [49]. However, population dynamics of species with short generation times reflect many biotic and abiotic factors and, together with impacts of abiotic factors on trap attractiveness, make such comparisons difficult. A different way is to experimentally alter the expected WT of a species and then examine the impact of this alteration on field performance. For example, within a generation, WT (or TSM) can be modified through acclimation and then compared in field studies on performance [50].

Another approach is to consider a group of species closely related to the individual target species and assess if a comparative analysis of the group using WT has any explanatory power in terms of relative species distributions and relative abundance in space and/or time. Chevalier *et al.* [51] found that WT weakly predicted stream fish population dynamics under changing climatic conditions, whereas trophic position, niche breadth, and their interactions were more important predictors. In *Drosophila*, simple measurements of thermal tolerance of species made under controlled conditions were linked to species distributions [5] and these have become increasingly sophisticated to include dynamic temperature changes [26]. However, such studies have not yet used WTs to predict dynamic changes in species distributions. Species that have narrow apparent WTs but occupy a similar range as those with wider WT would be of particular interest here for future monitoring, and can be identified from surveys [52]. In such comparisons, it will be important to consider any ongoing evolutionary adaptation affecting WTs (Figure 1) given that thermal limits can evolve [53,54].

A final powerful validation approach would be to link WTs to current and past changes in species distributions and population size, as has been done for the altitudinal retreat of stream fishes [55] and in comparisons of occupancy in past versus recent periods in bumblebee species [56]. There should be diverse opportunities for establishing such links given that climate change effects are leading to the extinction of many local populations [57]. Because of all the possible confounding effects on WT (Figure 1), attempts to validate them should consider groups of related species with similar behavioural and acclimation ability.

Concluding remarks and other perspectives

The literature on thermal vulnerability indices has expanded enormously in recent years. Here we have argued that while vulnerability indices will continue to be used (because of their simplicity and ease of application), we need to be aware of their limitations including the lack of validation. At present, there are too many outstanding questions to consider thermal vulnerability indices reliable. We point out that WTs will vary widely depending on a range of inputs including intrinsic factors such as the age and condition of the organism as well as extrinsic factors including the

Outstanding questions

Are there examples where WTs and TSMs can be validated as predictors of vulnerability across related species and what factors characterise these successful situations? The limited data available so far suggest that WTs appear to be successful in predicting distribution shifts in some species groups but not in others, and more tests are required before any general patterns can be derived. Such tests would need to be mindful of the details on obtaining useful metrics of upper thermal limits and quantifying the environmental variation in a manner described here.

What are the systematic biases when assessing thermal vulnerability indices of multiple species across timescales and regions? When applying the concept of WT and TSM to climate change vulnerability predictions, understanding and accounting for sources of variation in both sides of the WT and TSM equations is necessary since, in most cases, there is a specific comparison of interest; for example, life-stages, populations, species, assemblages or habitat types, or across time-points. Systematic biases can arise when for example, particular datasets (taxonomic groups or regions) are generated from protocols devised from logistical constraints or common practices rather than ecological relevance. Large scale compilations of WT and TSM data are required to test for these biases.

Are safety margins phylogenetically conserved? If yes, can we estimate safety margins of threatened species from (nonthreatened) close relatives? While upper thermal limits are sometimes found to be phylogenetically conserved, the effects are often weak, perhaps due to the often noisy compilation of diverse methods and metrics used in these large scale analyses. Further large scale systematic work, controlling for the aforementioned potential biases, in vulnerability estimates would be valuable especially in non-model taxa and from understudied geographic regions.

When and for which species or taxonomic groups do we advise managers (and funders) to follow a comprehensive and challenging mechanistic approach versus a thermal

nature of thermal stresses and prior conditions. These considerations will apply differently to the traits (particularly CT_{max} vs T_{opt}) included in the indices. For instance, the rate of temperature change will have a large impact on CT_{max} rather than T_{opt} . It is important to recognise that TSMs and WTs only measure a single facet of climate vulnerability. When assessing whether WTs can predict responses to climate change, they will need to be considered alongside other factors (e.g., water and food availability), including resilience or rescue factors, such as evolutionary processes and novel associations, and other types of traits (e.g., niche breadth and trophic position) and their interactions (Table 1).

Apart from thermal vulnerability indices based on physiological traits, there are other ways of determining the likely safety of populations into the future. One possible approach is to ignore WTs altogether and assume that the current abiotic set of factors that sets the limit of an organism's distribution already represents its limit. In this case, a species should not have existed in a particular place in the past (in the absence of evolution) when current distributions are hindcasted against paleoclimates during periods when climatic changes were evident, such as during Pleistocene glacial cycles [58]. Current limits might be compared against past species distributions based on fossils [59] supplemented by DNA-based assessments of tissues from living organisms [60] and/or well-documented changes in recent climate [56]. Any discrepancies between predicted and well-documented past distributions may reflect WTs, particularly if tolerance limits derived from past species distributions over-predict their current distributions. However, this type of information may not be available for many species and the issues of microclimatic variation, behavioural responses and biotic interactions will also need to be considered. Ongoing phenotypic changes within lineages and the extent to which past climatic changes were tracked through migration will impact predictions [61], and these will also need to be assessed simultaneously. Again, molecular data can help in assessing migration patterns in addition to changes in population size that will influence evolutionary adaptability. As well as being derived from comparisons in time, WTs can be assessed by comparisons across space when animals are deliberately or accidentally introduced into new regions (see Outstanding questions). The data collected so far suggest that climatic niches in these instances are largely conserved [62], whereas range expansions might be expected in the case of large WTs.

vulnerability index approach when assessing species' vulnerability to climate warming? A WT and TSM approach serves as a good 'broad brushstrokes' starting point in the absence of detailed organismal or environmental information, but it can be quickly surpassed with focused effort on acquiring specific information about the species' biology and environmental conditions. Understanding when that research effort versus information gained trade-off is no longer worthwhile for the breadth and depth of insights gained will be paramount to managers and policy makers.

Acknowledgements

We would like to thank Jessica Allen for preliminary literature searches and Art Woods for useful comments on the manuscript draft. We thank the Editor and three anonymous reviewers for their constructive comments.

Declaration of interests

None declared by authors.

Supplemental Information

Supplemental Information associated with this article can be found online at <https://doi.org/10.1016/j.tree.2021.07.001>.

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